SOP Number:	**1**	Revision Number:	**2**	Effective Date:	**7/18/2025**
Title:	**Purification of Molten Fluoride Salts by Gas Sparging**				
AUTHORIZATION					
Created by (Printed Name):	**Kyle Williams/Nathaniel Thomas**		Title: N/A	**Graduate Research Assistant**	
Created by (Signature):				Date:	
Approved by (Printed Name):			Title:		
Approved by (Approval – Signature):				Date:	

1. Purpose

The purpose of this SOP is to outline the steps involved in the purification of FliNaK at the D. L. Houston building.

2. Responsibilities

The Laboratory Manager will approve this SOP. The PI and Laboratory Manager will ensure that all personnel performing the procedures will be trained on the contents of this SOP. All researchers will ensure that materials to be transported to the facilities for analysis are packed in accordance with applicable Environmental Health and Safety procedures as well as local, state, and federal regulations. A researcher is either a student or a full-time staff research engineer working on a specific project within room 119 of the D. L. Houston Building. All personnel involved in the procedures are required to read and follow this SOP.

3. Training

The researcher must receive General Radiation Safety Training, respirator training, and task specific training on the contents of this SOP. The PI and Lab Manager will ensure that the general radiation safety awareness training and respirator training has been completed and signed by the researcher.

4. Equipment and Materials

4.1. Personal Protective Equipment

- Half-face respiratory with HF Filters

4.2. Gas Sensors

- Honeywell Satellite XT Gas Detector with HF Sensor
- Cosmos PS-7 with Hydrogen Sensor

4.3. Safety Equipment

- Fume Hood 119D

4.4. Heating

- ATS 3110-8.25-16-16 tube furnace with setpoint control
- HF Cylinder heating band
- Grainer Samox heating tape
- Alumina Firebricks
- Kaowool insulation

4.5. Gases/Gas sources

- Peak Scientific Trace 1200 Hydrogen Generator
- HF cylinder
- Arcylinder
- H_2

4.6. Hardware

- C-276 Reaction vessel and lid
- Ni-200 Reaction vessel liner
- 316SS Transfer Vessel
- 304SS Vacuum Chamber
- 50 micron 316SS filter
- 20 psi Hastelloy C-276 Burst Disk
- Manual SS Sapphire-Sealed Variable leak valve
- HF Flow Controller
- AALBORG He Mass Flow Controller
- AALBORG H_2 Mass Flow Controller
- Hf Flow Controller
- HF Regulator

4.7. Analytical Equipment

- SRS Residual Gas Analyzer (RGA) 100
- NI Data Acquisition Unit
- Scale
- Laptop

4.8. Vacuum Equipment

- Ionization gauge
- Edwards RV5 2 Stage Rotary Vane Pump
- Leybold Turbovac 90i Turbomolecular Pump

4.9. Chemicals

- 500 mL Teflon Bottle with CaO
- 2.5 gal HDPE Carboy with KOH solution
- Ultra-high purity water
- CaO solid
- KOH solution

- FLiNaK

5. Procedure

5.1. General Information

This section below covers general information for salt purification from development including user safety, sample records, waste disposal, and cleaning.

5.1.1. Personal Protective Equipment and Hazardous Materials Handling

a) Due to the use of hydrogen fluoride and fluoride salts(HF) all users must be authorized and be fitted for respirators before handling materials and conducting tests.
b) Appropriate lab attire must always be worn.
c) Safety goggles/glasses must be worn when working with materials outside of the glove box.
d) Users should be aware of the hazards when handling hydrogen fluoride/fluoride salts and associated possible health effects.

5.1.2. Sample Identification and Labeling

a) All samples need to be labeled and placed into sample vials with lid attached securely. Sample labels must contain salt information, additives, waste from processing technique, mass, date, unique identifier, and user initials.

5.1.3. Records

a) A laboratory notebook will be kept with all activities and dates recorded.
b) A record will be kept of test vessel cleaning that identifies each vessel and lid, last cleaning date, fluorine content, and pH recorded after the cleaning process.
c) All RGA data will be stored in spreadsheets.

5.1.4. Users

a) Due to the use of HF all users must be authorized and be fitted for respirators before handling materials and conducting tests.
b) Users should be aware of the hazards when handling HF and associate possible health effects.

5.1.5. Waste Disposal and Cleanup

a) Four primary waste disposal containers will be in place.
   i. A bottle will be supplied for liquid waste.
   ii. A container will be supplied for solid waste.
   iii. A bottle/container will be supplied for waste generated during reaction and transfer vessel cleaning.
b) All work surfaces should be left clean and orderly after use.
c) Samples must be stored with appropriate labels and lids tightly secured.
d) Samples must be stored in a glove box to prevent contamination.
e) All appropriate documentation must be filled out.

5.2. FLiNaK & FLiBe Salt Production

Currently all FLiNaK and FLiBe salt is purchased from a supplier. This purchased salt is already fused into its eutectic and only requires purification. If the need to produce either salt from their constituent compounds presents itself, this SOP will be revised to include this process.

5.3. System overview

This is an extensive and complex system with many components, control elements, hazards, sensors, and other pieces of equipment. An overview of the outside of the fume hood is shown in **Figure 1**. Outside of the fume hood is the following:

1. An argon gas cylinder with a regulator and shutoff valve attached.
2. An argon feed line that runs into the ceiling and back out of the ceiling next to the fume hood
3. A laptop which runs several control and analysis programs for the system
4. A hydrogen generator, alongside a bottle of ultra-high purity water
5. A rotary vane vacuum pump, with the outlet attached to a water trap
6. A DC power supply for running various pieces of equipment inside the fume hood
7. A heating tape controller.
8. A vacuum gauge
9. A NI Data Acquisition Assistant

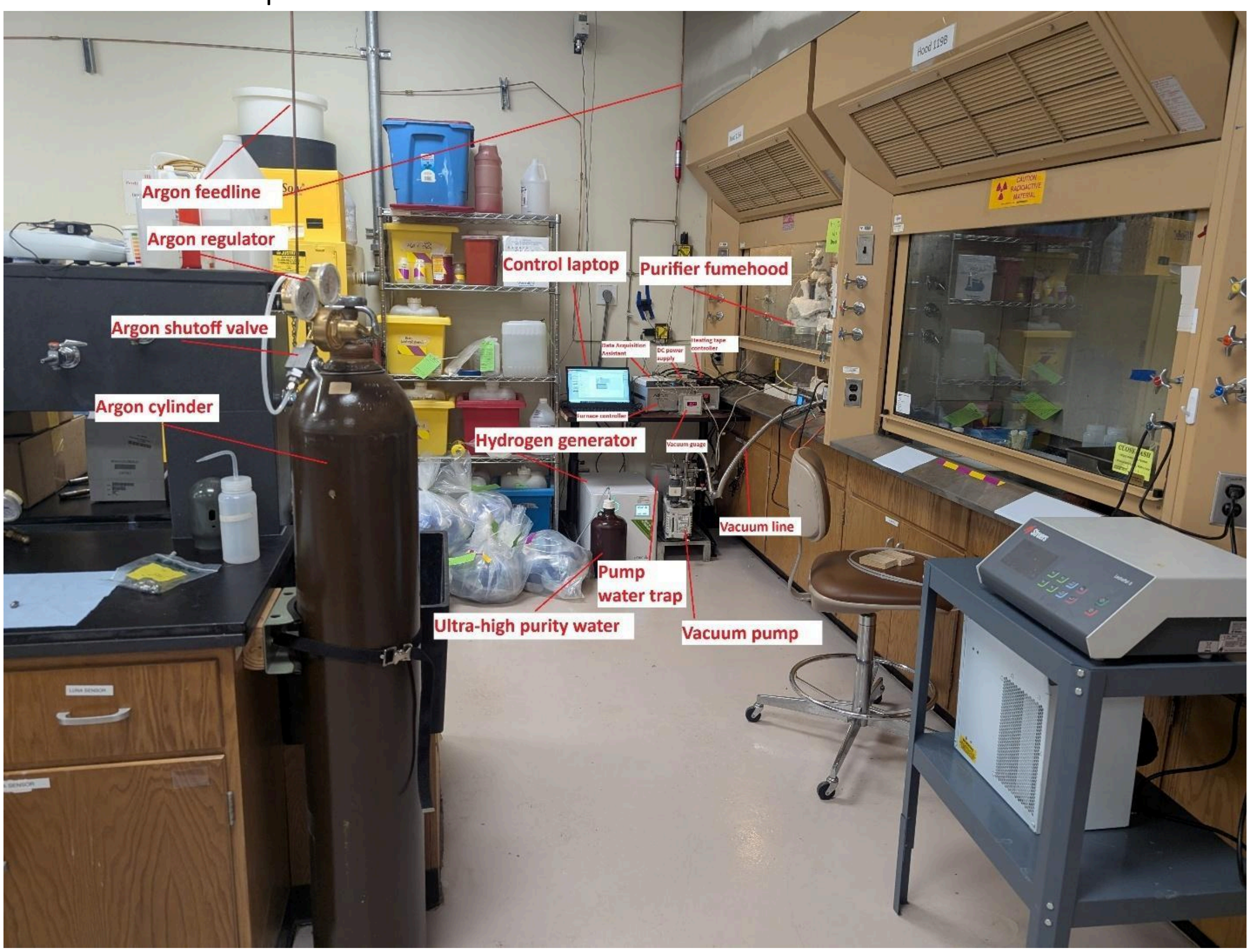


Figure 1: *View of the purification system outside of the fume hood.*

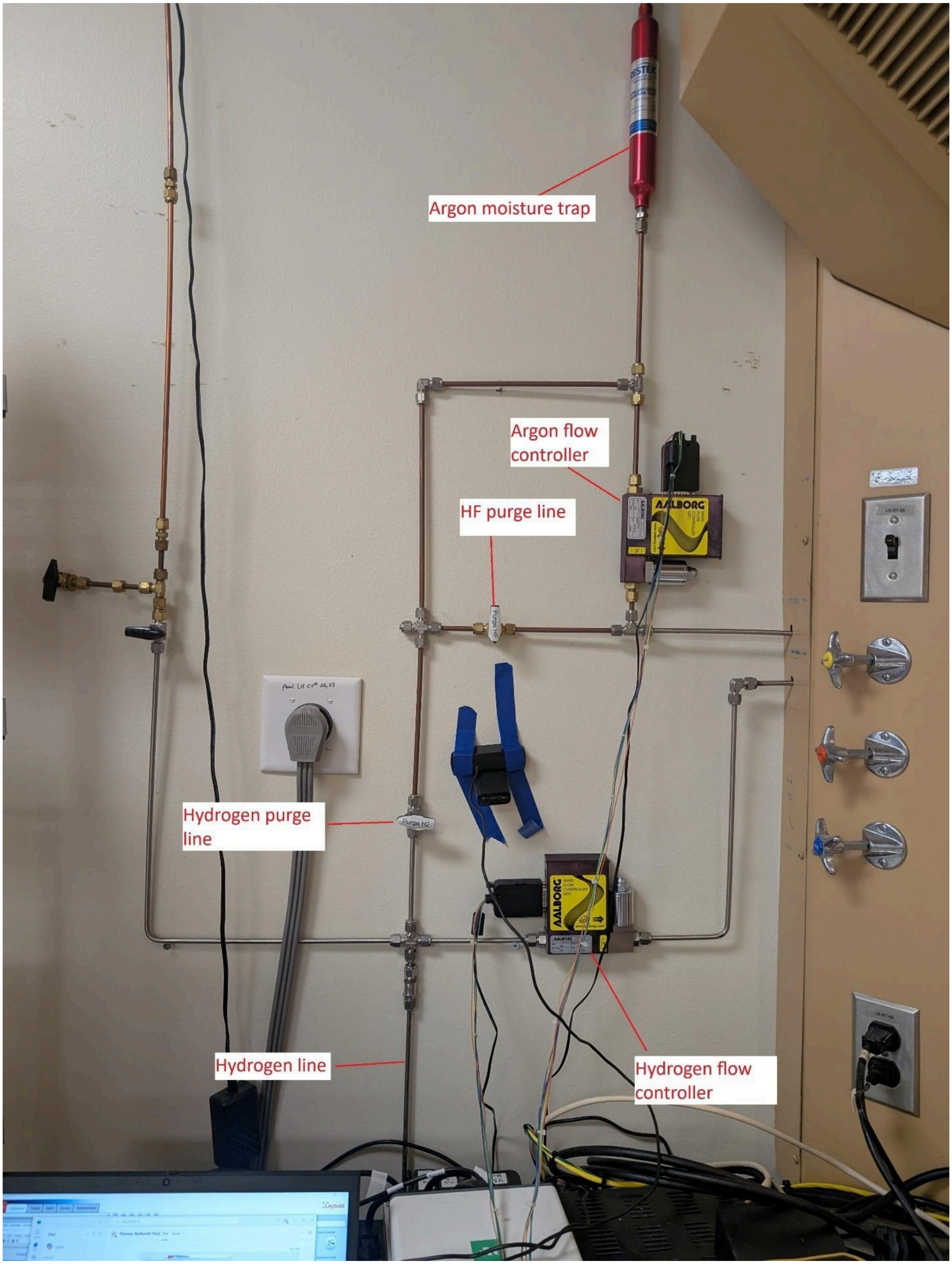


Figure 2: Gas lines for the purifier system

Figure 2 illustrates the gas lines for the purifier system. It consists of a hydrogen and argon flow controller, as well as feed and purge lines, along with control valves for several lines within the system.

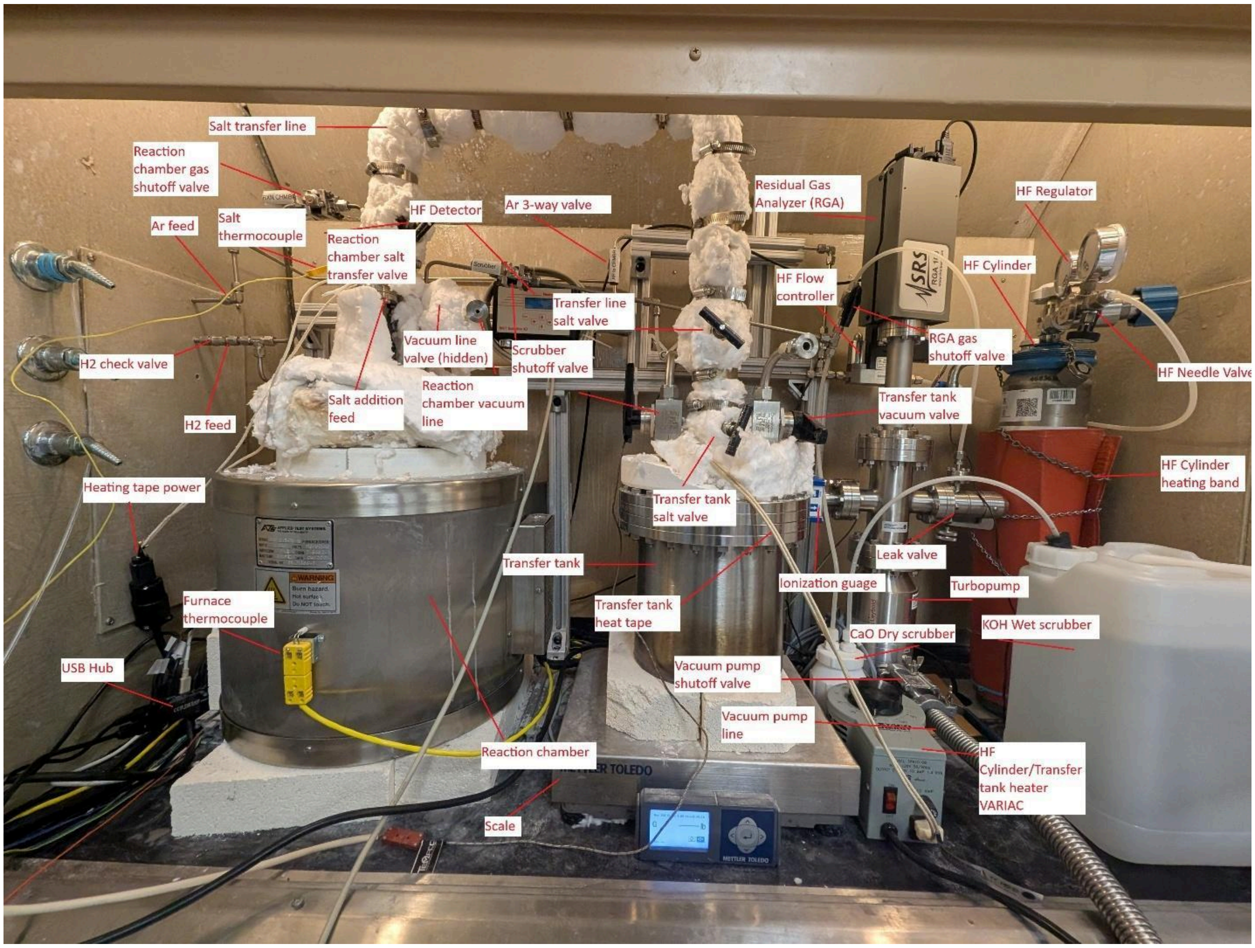


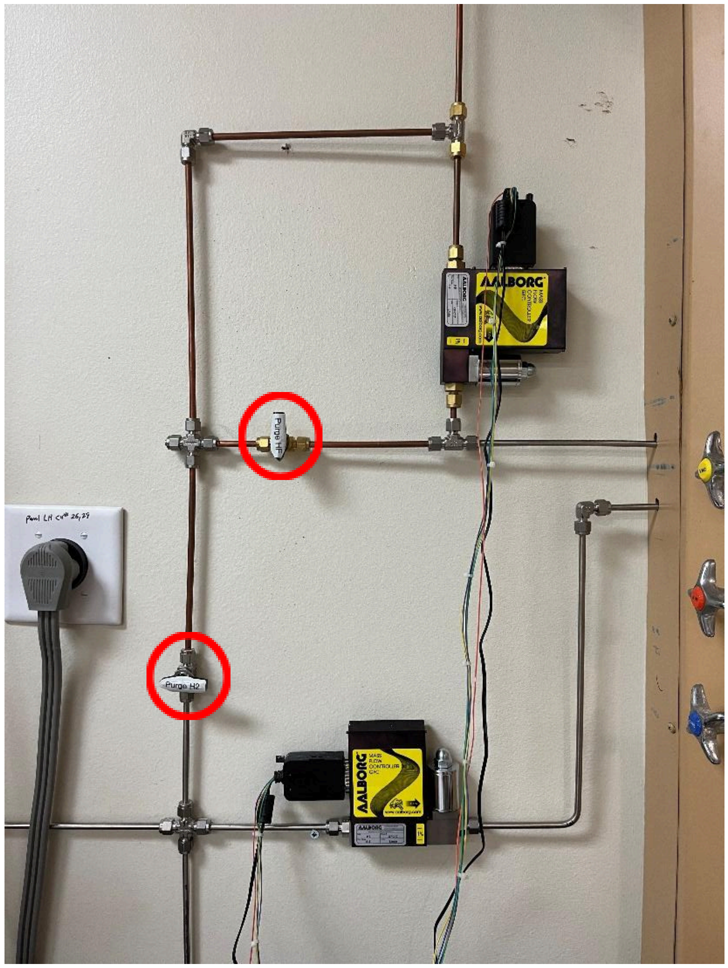
Figure 3: Purifier fume hood setup

Inside the fume hood, many components are present. Figure 3 is a largely comprehensive annotated image of the fume hood with each significant component labeled.

5.4. Startup

Due to semi-autonomous design considerations, vessels must be connected, and various instruments must be activated before the reaction can begin. This section covers the proper preparatory steps to be taken to ensure proper conditions for purification.

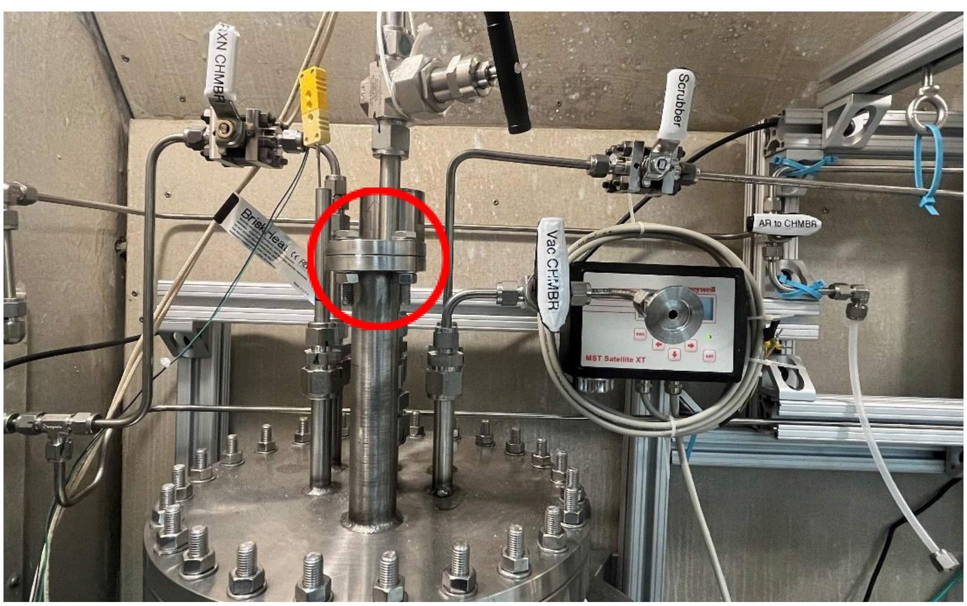
5.3.1. Ensure all gas cylinders contain enough gas for a full run of the process.
5.3.2. Open the valve on the Ar cylinder, regulator, and regulator valve.
5.3.3. Close both the “Purge HF” and “Purge H2” valves (shown below). *Closed is indicated if the valve handle is parallel to the direction of flow.*


AALBORG
Purge H2
AALBORG

5.3.4. Attach a cleaned transfer tank to the end of the salt transfer line and Ar purge line via the Swagelok fittings shown below. Make sure the transfer vessel is sitting on fire brick insulation, and that the scale below the vessel is recording the weight.

5.3.5. Either zero the scale beneath the transfer vessel or record the weight of the empty vessel before proceeding.

5.3.6. Loosen and remove the bolts on feed tube of the reaction vessel shown below and remove the lid.

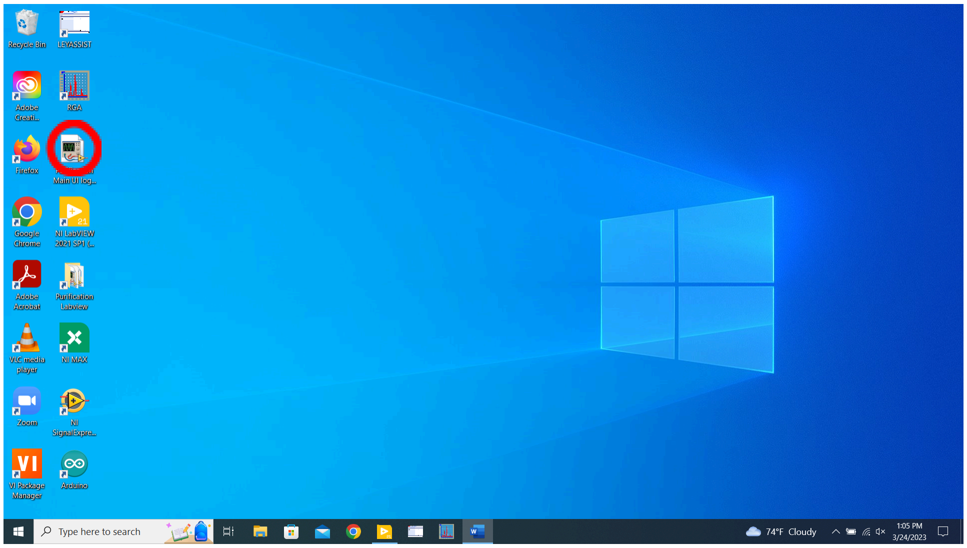


5.3.7. Weigh out approximately 10 kg of either FLiNaK or FLiBe salt within a glovebox and quickly add it to the reaction vessel. Ensure all salt is within the liner of the vessel.

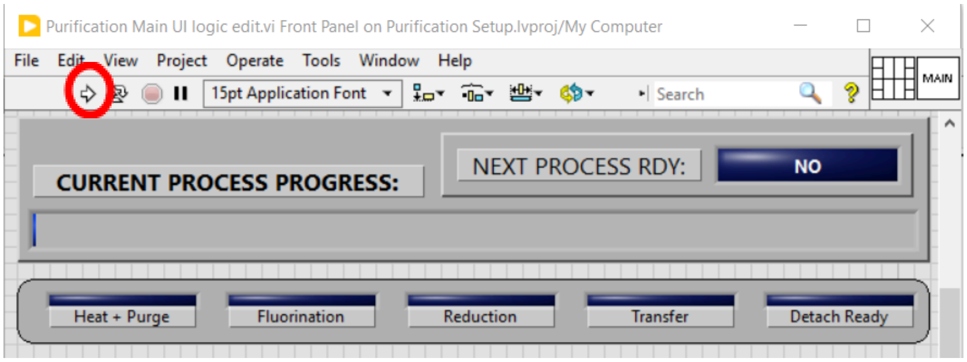
5.3.8. Place a new sealing gasket and the lid back on the feed tube and bolt them down to ensure an airtight seal.

5.3.9. Open the "RXN CHMBR" valve and "Scrubber" valve (shown below) and ensure the "AR to CHMBR" valve is pointing up. O*pen is indicated if the valve handle is parallel to the direction of flow.*

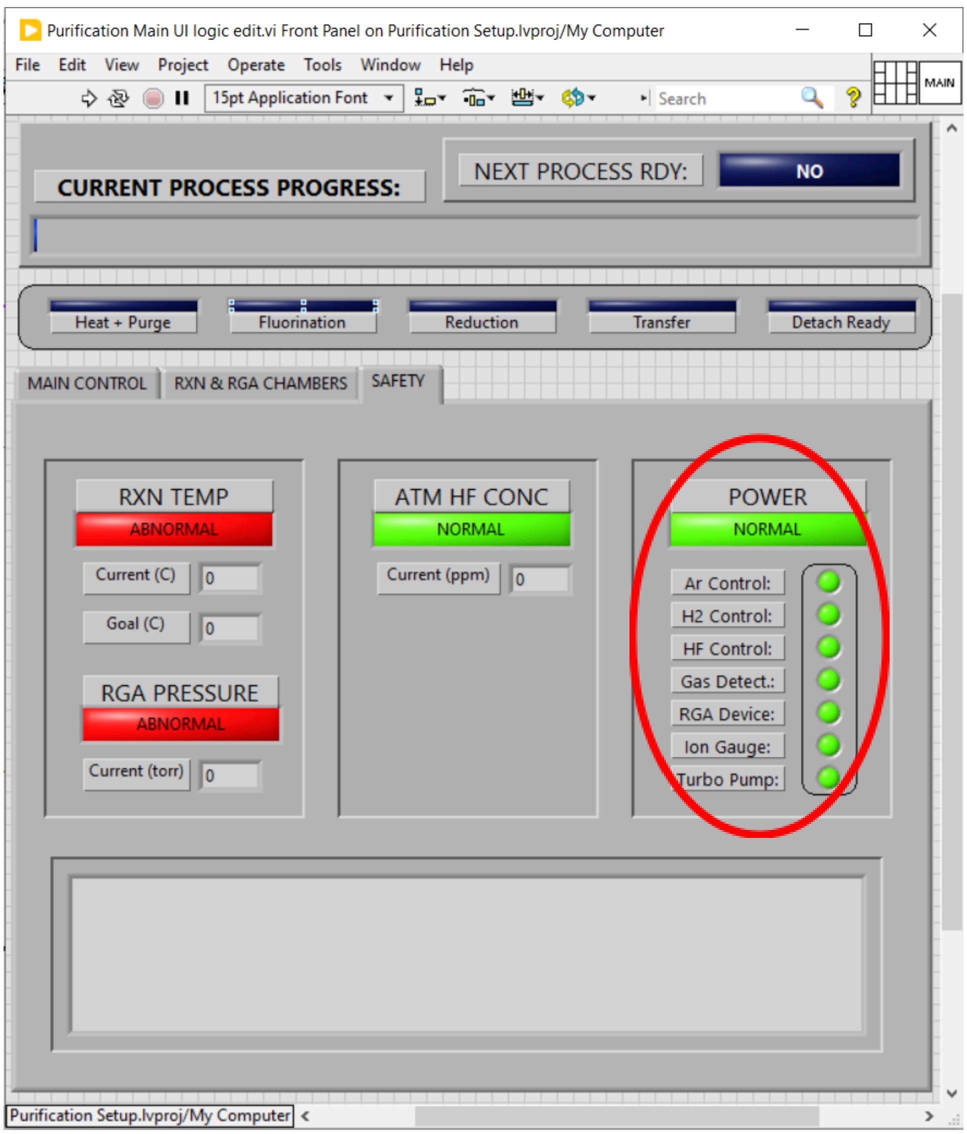
5.3.10. Open the Purification Main UI VI (shown below).



5.3.11. Click the run button (shown below).

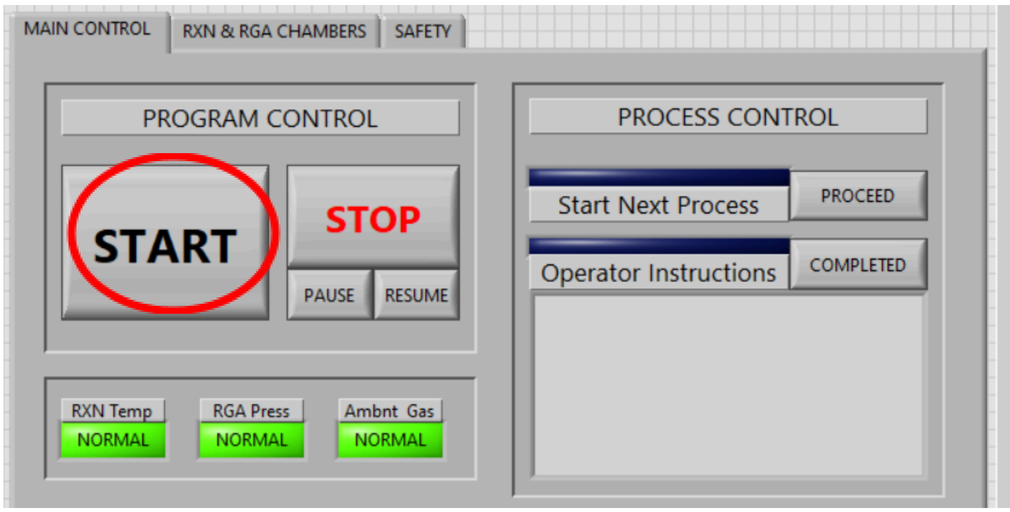


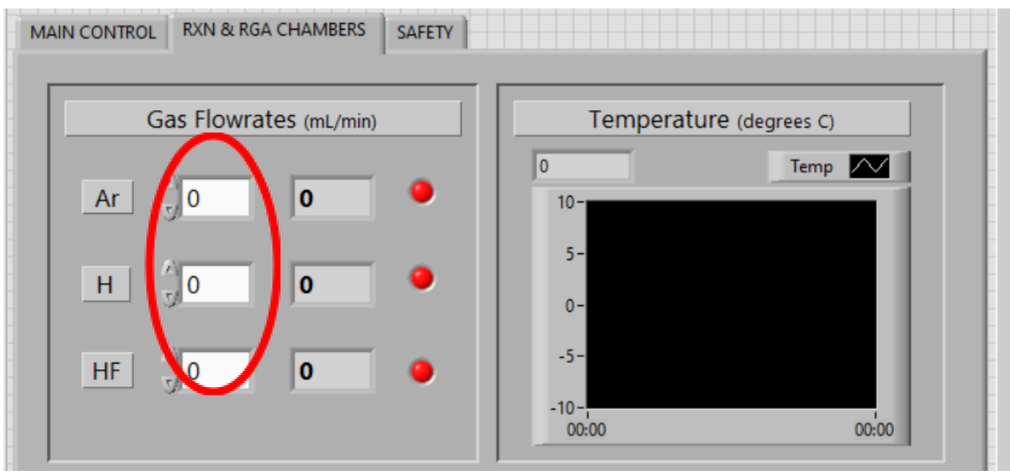
5.3.12. Check the “SAFETY” tab to make sure all necessary devices are on (shown below).



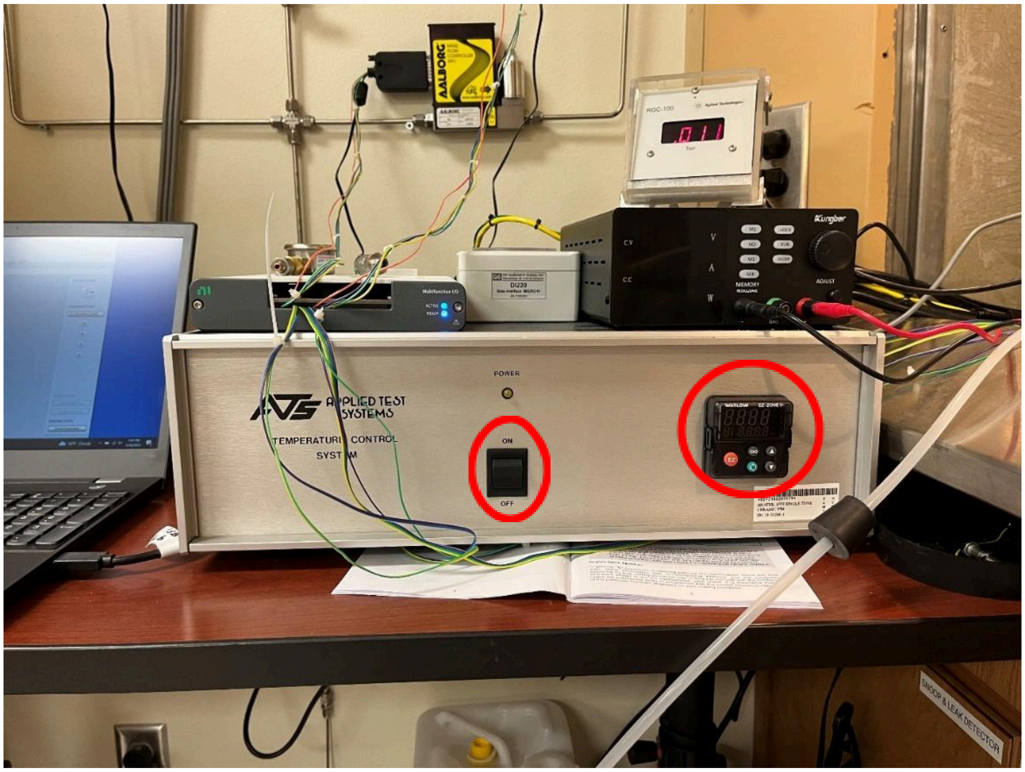
5.3.13. Select an operation mode from the dropdown menu (shown below).

5.3.14. If “Operator” was selected, then press the start button (shown below) to begin the “Heat + Purge” stage. If “Engineer” was selected, then set the argon flow to at least 500 mL/min from the “RXN & RGA CHAMBERS” and wait several minutes before proceeding.



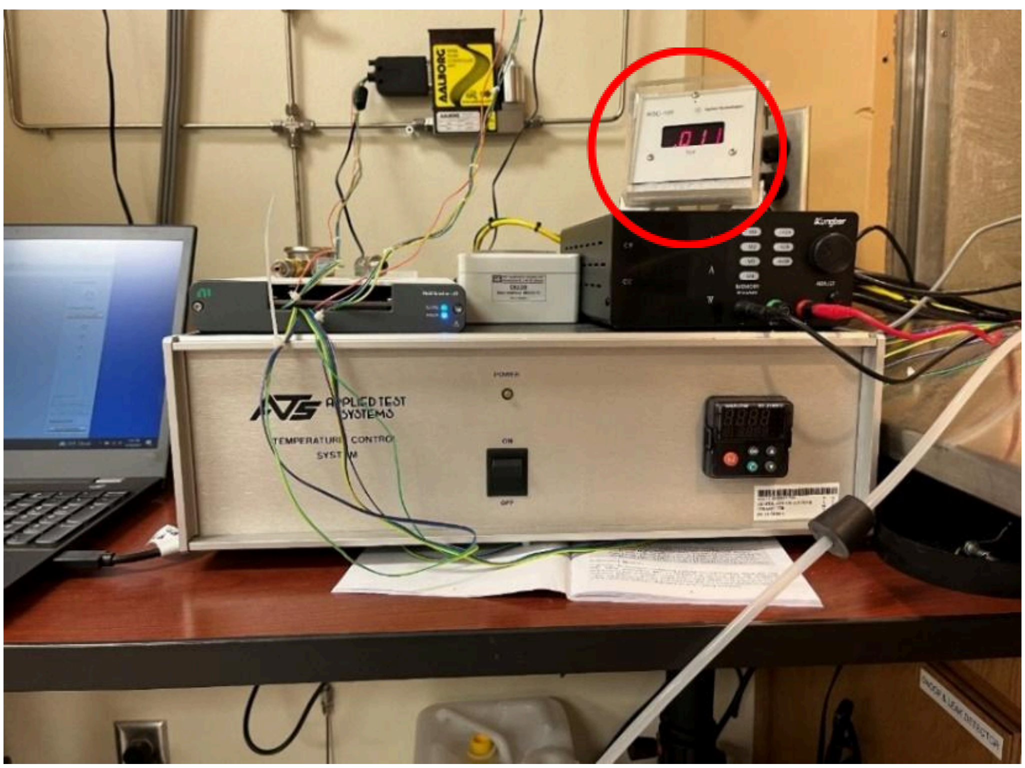


5.3.15. Turn on the furnace and change the setpoint to 650°C using the Watlow controller (shown below).

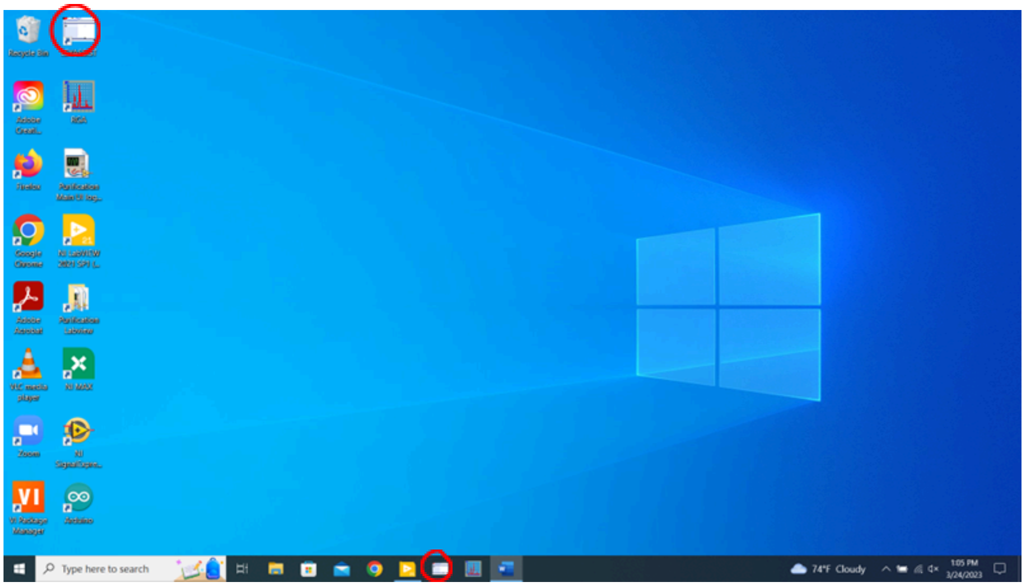


5.3.16. Turn on the roughing pump by pressing the switch on the side of the pump (shown below).

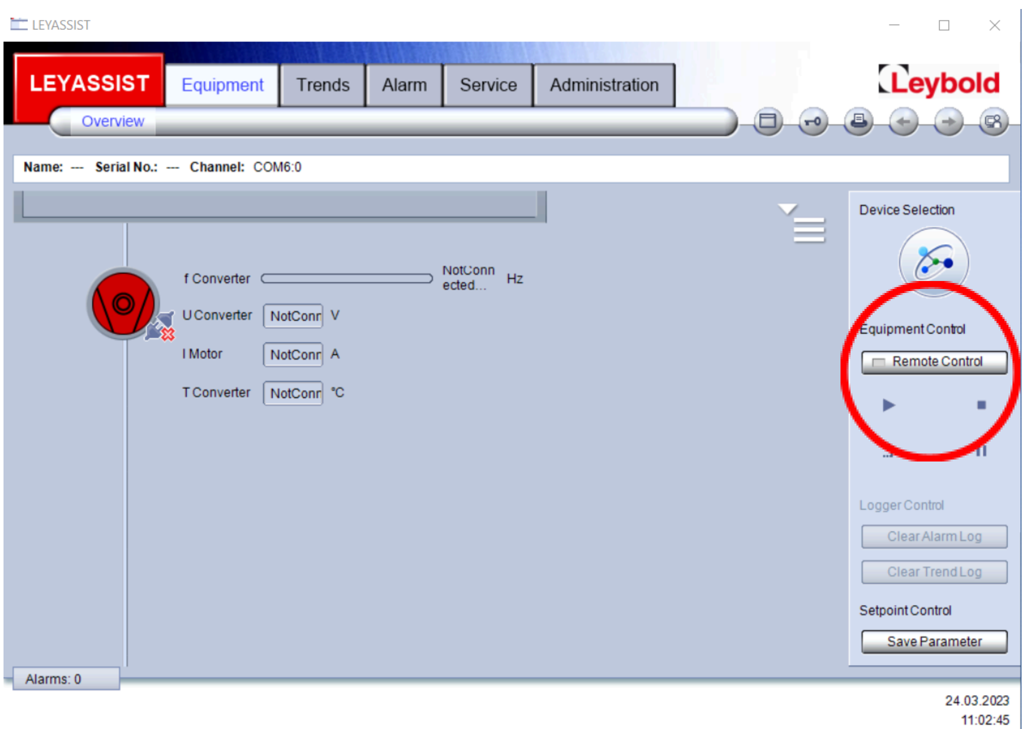
5.3.17. Open the vacuum line valve (shown below) and wait until the low vacuum gauge displays a value of 0.013 or less.



5.3.18. Turn on the turbo pump using the Leyassist application as shown below.



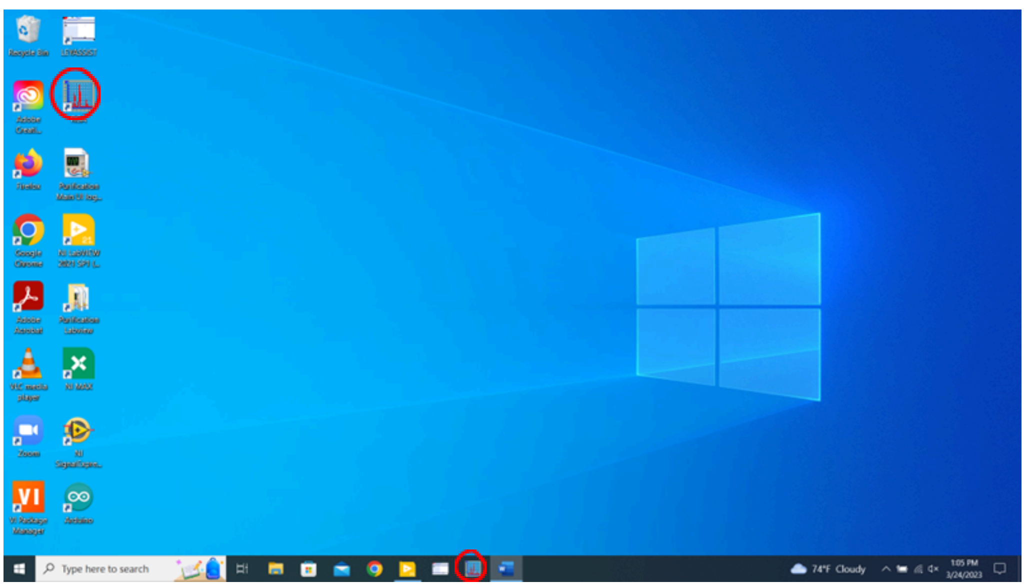
5.3.19. Press the “Remote Control” button followed by the play button to start the turbo pump.



5.3.20. Turn on the ion gauge (shown below) by pressing the menu button and then the enter button and wait until it displays a value below 1E-6.

5.3.21. Open the RGA chamber valve (shown above) and ensure that the pressure doesn't rise above 1E-5 torr. This may require adjusting the RGA Fine valve.

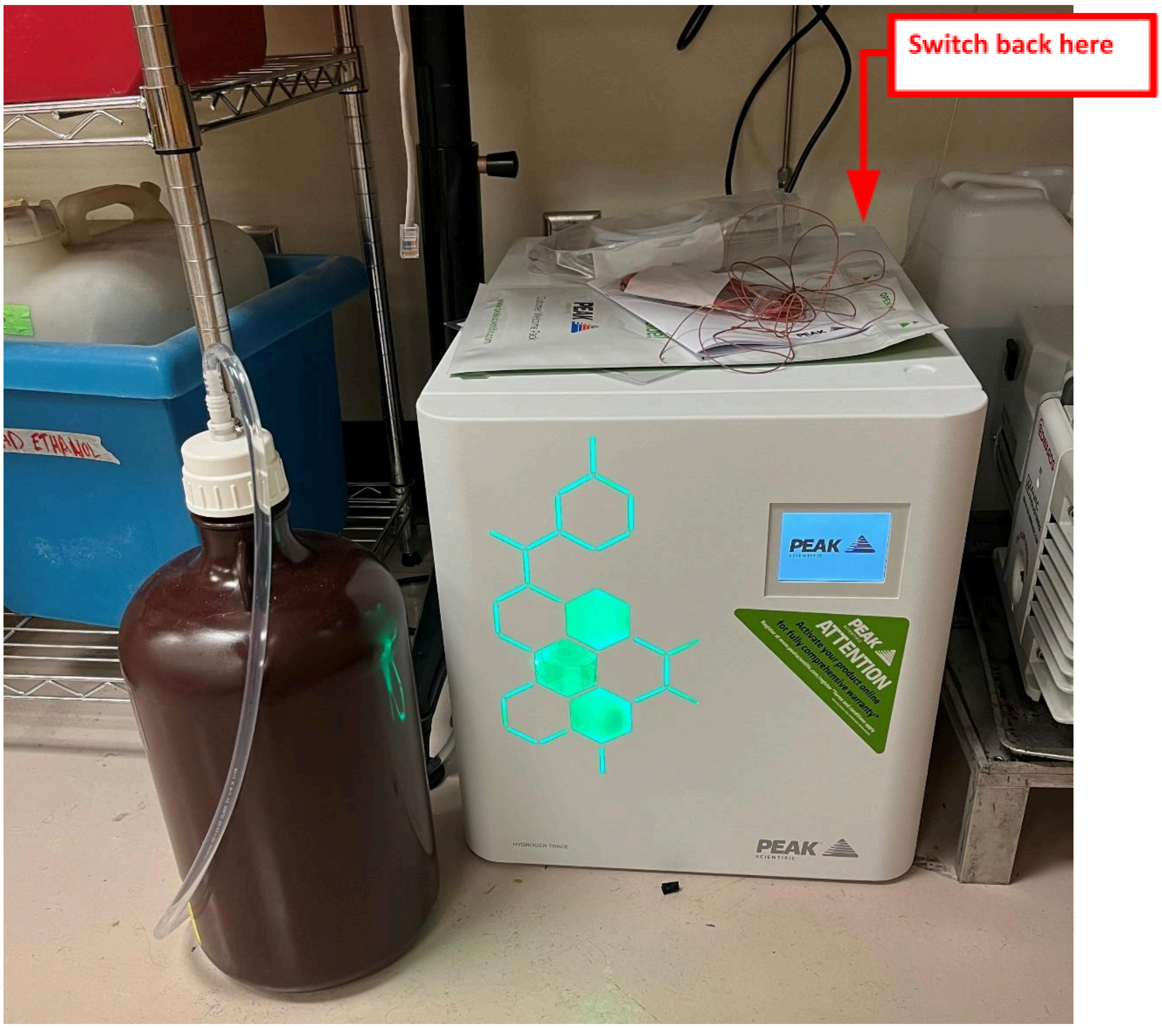
5.3.22. Turn on the RGA by opening the RGA application (shown below).



5.3.23. Ensure that the water bottle to the H2 generator isn't empty. If it needs to be topped off or refilled, use only ASTM Type II deionized water.

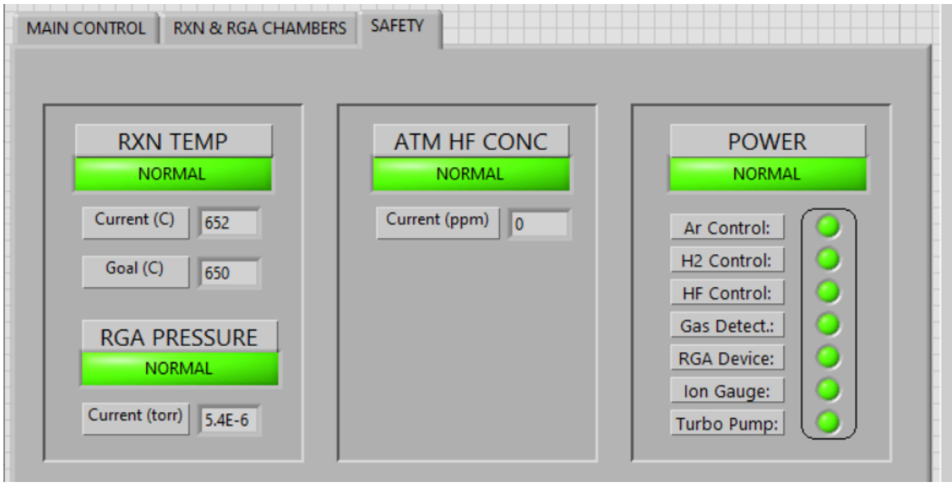
5.3.24. Turn on the hydrogen generator by flipping on the switch on the back of the unit (indicated below).

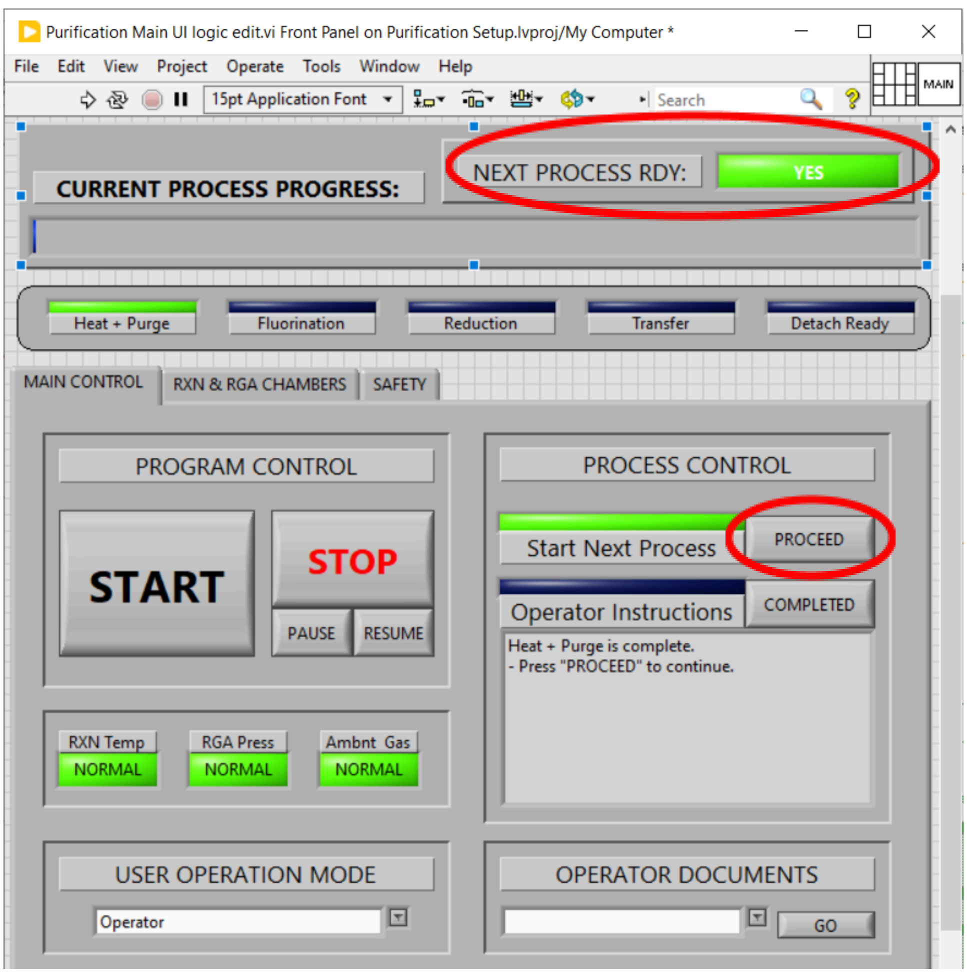
5.3.25. Allow the unit to go through startup and press the start button on the screen. Should the unit fail to properly startup, contact the PI.



5.3.26. The PI will set up the HF cylinder to minimize the operator interaction. All the operator should need to do is open the needle valve on the regulator (shown below). If the HF cylinder is not properly setup, contact the PI.

5.3.27. Once all operator tasks have been completed, the safety tab shows all systems are good to go, and the “NEXT PROCESS RDY” indicator is active (shown below), press the “PROCEED” button (shown below) to begin gas sparging. For Engineer mode, this will be disabled, and the gas flow must be adjusted in the “RGA & RXN CHAMBERS tab.



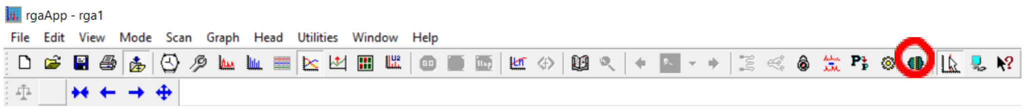


5.4. Gas Sparging

Purchased salt or salt produced in-house will contain oxygen, sulfur, and metallic impurities. The method chosen by the for this project is gas sparging with a fluorinating gas mix. This fluorinating gas is often diluted with H_2 gas to prevent/manage the corrosion of reaction vessels. The use of HF diluted to a 1 to 10 ratio in H_2 has proven effective in past literature and will be used. This fluorination step is followed by a reduction step. Sparging with H_2 for many hours was chosen to fulfill this step. Reductive metal and hydride additions and/or electrochemical acceleration may be used in future experiments to speed up the reductive step. Should this occur, this SOP will be revised to include procedures for those additions. To ensure operator safety, the PI is the only one that will handle any of these chemicals directly.

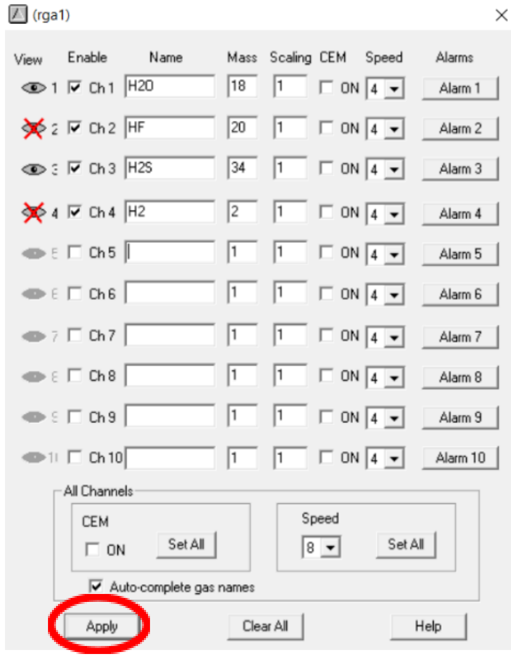
5.4.1. The program will give operator instructions to ensure the RGA chamber preparation has been completed. If it hasn't, complete 5.3.16 - 5.3.22 before proceeding.

5.4.2. If it isn't already, ensure that he RGA is connected by clicking the Open Connector List icon (shown below) in the RGA App.



5.4.3. Put the graph into Pressure versus time mode by clicking the icon (shown below in the RGA App.

5.4.4. Click the View current scan options icon (shown below) to adjust the graph display settings and adjust the graphing settings to display for water (mass=18) and hydrogen sulfide (mass=34).



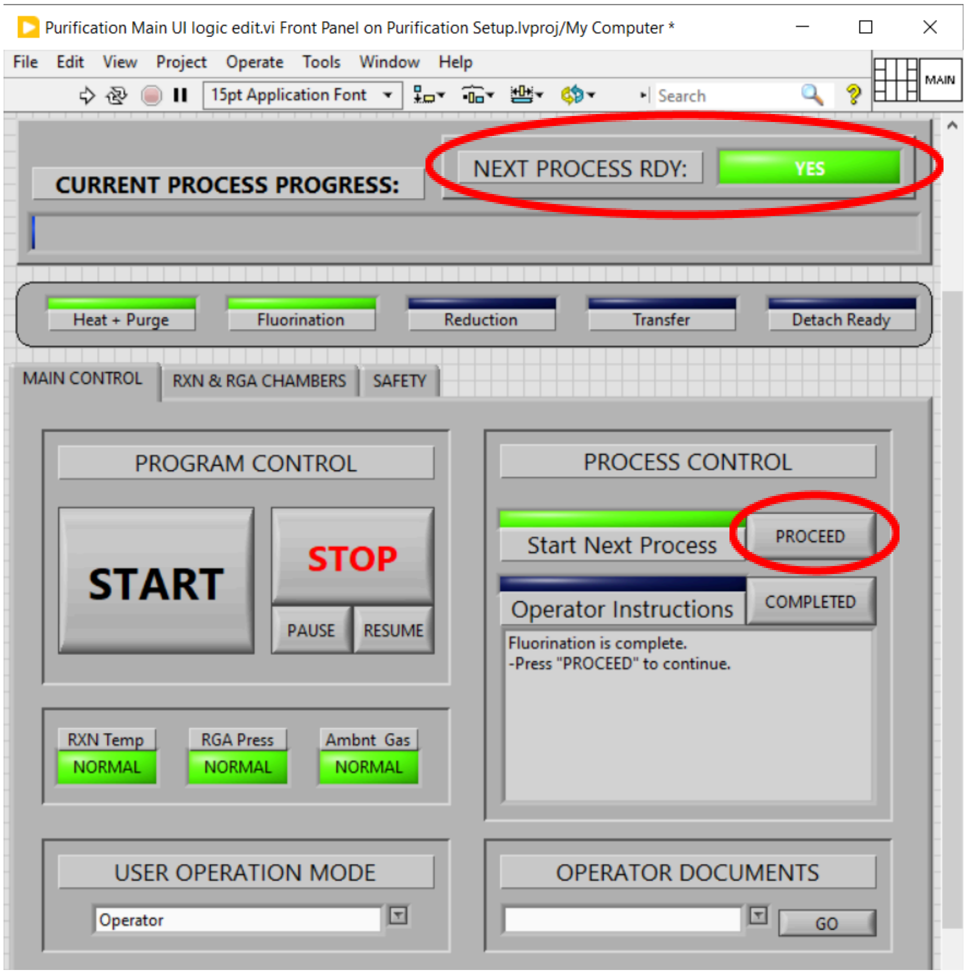
5.4.5. Open the RGA filament by clicking the button (shown below) in the RGA App.

5.4.6. Start scanning with the RGA by clicking the button (shown below) in the RGA App.

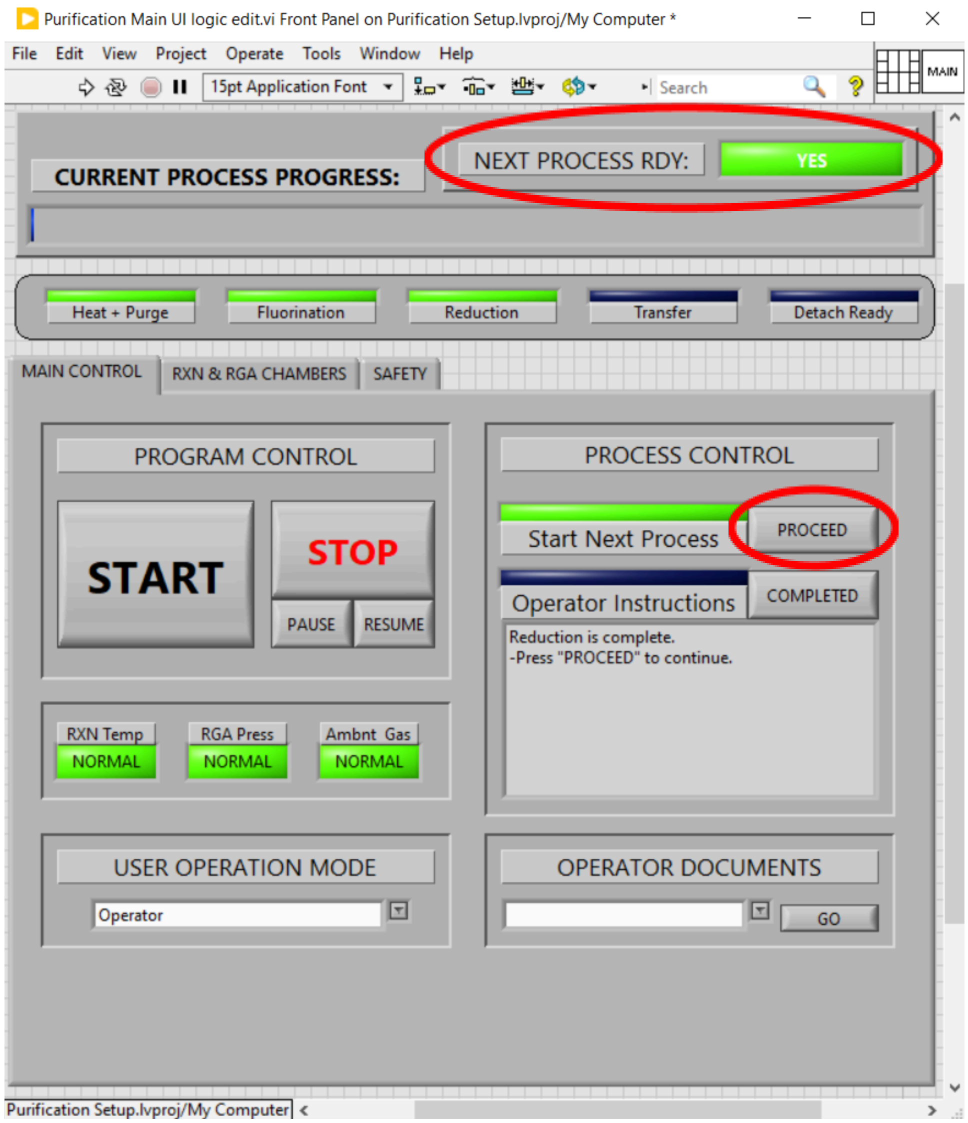
5.4.7. Once all previous operator steps have been completed, the system will begin fluorinating. It is currently programmed to fluorinate for 24 hr. Should the RGA chamber pressure get too high, it will pause the fluorination until pressure is brought back down.

5.4.8. Once the “NEXT PROCESS RDY” indicator is active (shown below), save the scan data from the RGA, then create a new chart displaying the HF and H2S pressures over time, like steps 5.4.3 and 5.4.4.

5.4.9. Press the “PROCEED” button (shown below) to begin Reduction. For Engineer mode, this will be disabled, and the gas flow must be adjusted in the “RGA & RXN CHAMBERS tab.


Purification Main UI logic edit.vi Front Panel on Purification Setup.lvproj/My Computer *
File Edit View Project Operate Tools Window Help
15pt Application Font
Search
MAIN
CURRENT PROCESS PROGRESS:
NEXT PROCESS RDY:
YES
Heat + Purge
Fluorination
Reduction
Transfer
Detach Ready
MAIN CONTROL
RXN & RGA CHAMBERS
SAFETY
PROGRAM CONTROL
START
STOP
PAUSE
RESUME
PROCESS CONTROL
Start Next Process
PROCEED
Operator Instructions
COMPLETED
Fluorination is complete.
-Press "PROCEED" to continue.
RXN Temp
NORMAL
RGA Press
NORMAL
Ambnt Gas
NORMAL
USER OPERATION MODE
Operator
OPERATOR DOCUMENTS
GO

5.4.10. The program will now reduce the molten salt with H_2 for 48 hr. Should the RGA chamber pressure get too high, it will pause the fluorination until pressure is brought back down.
5.4.11. Once the "NEXT PROCESS RDY" indicator is active (shown below), save the scan data from the RGA, turn off the filament, and close the RGA App.



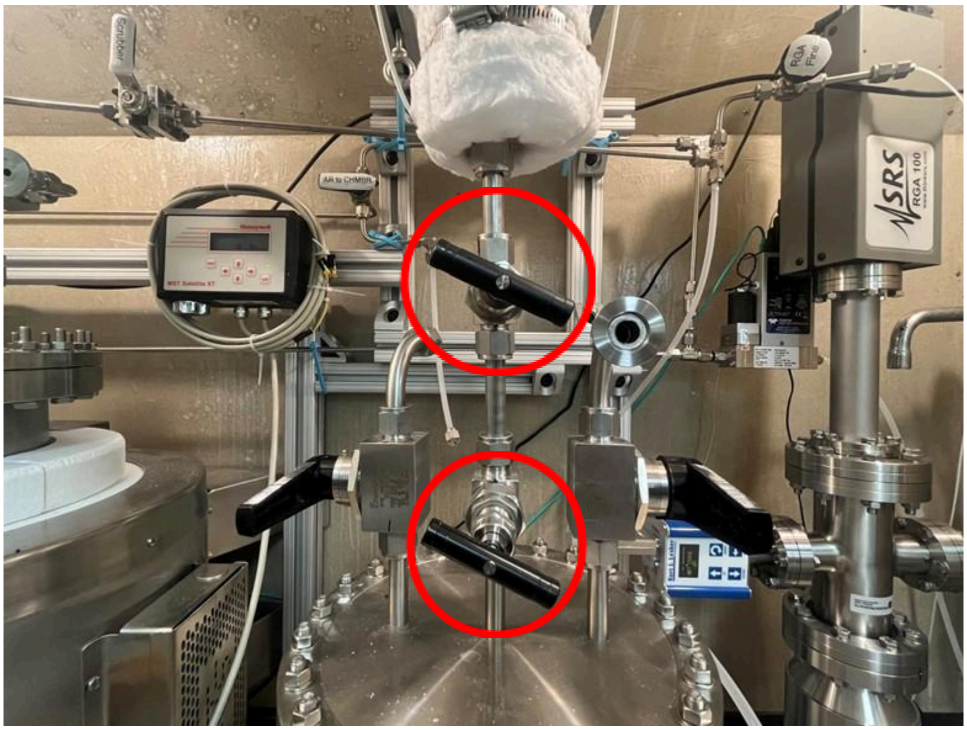
5.4.12. Close the RGA chamber valve and turn off the ion gauge by pressing the menu, down arrow, then enter buttons.
5.4.13. Turn off the turbo pump, either by turning off remote control from the Leyassist app, or by clicking the "Off/On" button in the "RXN & RGA CHAMBERS" tab.
5.4.14. Close the vacuum line valve and turn off the roughing pump.
5.4.15. Press the "PROCEED" button (shown above) to begin Transfer. For Engineer mode, this will be disabled, and the gas flow must be adjusted in the "RGA & RXN CHAMBERS tab.

5.5. Salt Transfer

The purified salt must be moved and stored in inert atmosphere to maintain purity. The most effective way to do this is to either seal and transfer the reaction vessel or transfer the salt while still molten into an inert atmosphere container. To minimize the disassembly of the reaction vessel, to maintain its seals, molten salt transfer was chosen.

5.5.1. Close the Scrubber valve.

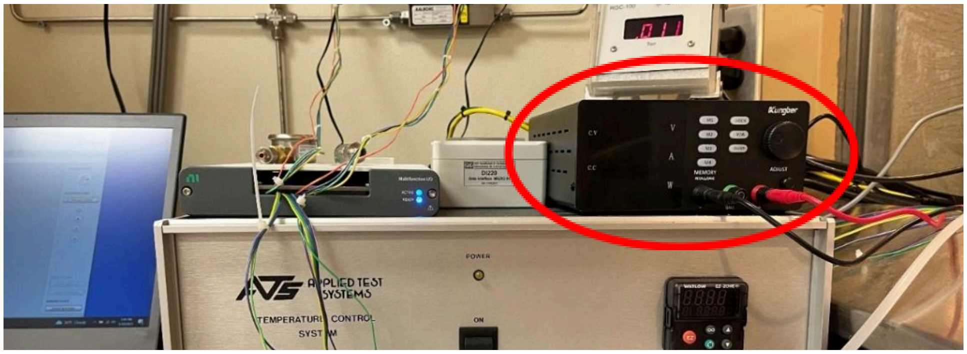
5.5.2. Open the 2 transfer line valves (shown below).



5.5.3. If transfer line and vessel need to be purged, follow the steps below:

a) Close the vacuum line valve and disconnect the roughing pump hose from the vacuum line valve.
b) Connect the roughing pump hose to the vacuum fitting on the transfer vessel and turn it on.
c) Connect the line below the “AR to CHMBR” valve to the Swagelok fitting on the transfer vessel and set the “AR to CHMBR” valve to the down position.
d) Open the “HF Purge” valve on the wall to the left of the fume hood.
e) Alternate opening for 30 seconds and closing the ball valves on the transfer vessel lid several times, starting and finishing with vacuum.
f) Close the “HF Purge” valve on the wall to the left of the fume hood.
g) Set the “AR to CHMBER” valve to the up position.
h) Turn off the roughing pump and reconnect the rouging pump hose to the vacuum line valve.

5.5.4. Turn on the heat tape power supply (shown below) and wait 10 minutes to allow the transfer line to heat up.



5.5.5. Open the last needle valve on the salt transfer line.
5.5.6. Observe the scale value and click the “COMPLETED” button when the mass indicated by the scale is no longer going up rapidly. This may take several minutes.
5.5.7. Record the mass of salt collected in the transfer vessel.
5.5.8. Turn the Power supply to the heat tape off.
5.5.9. Close all the transfer line needle valves.
5.5.10. Press the “PROCEED” button (shown below) to begin Detach. For Engineer mode, this will be disabled, and the gas flow must be adjusted in the “RGA & RXN CHAMBERS tab.
5.5.11. Ensure that the transfer vessel has cooled before attempting to disconnect or move.
5.5.12. Loosen the lower Swagelok as fitting shown below to disconnect the transfer vessel from the transfer line (shown below).

5.5.13. Put the transfer vessel into an inert, dry atmosphere glovebox before opening.

5.6. Shutdown

This section covers the steps necessary to return the system to the state it would be at in the beginning of the 5.3 Startup section, thus reducing confusion and chance of improper startup for the next operator.

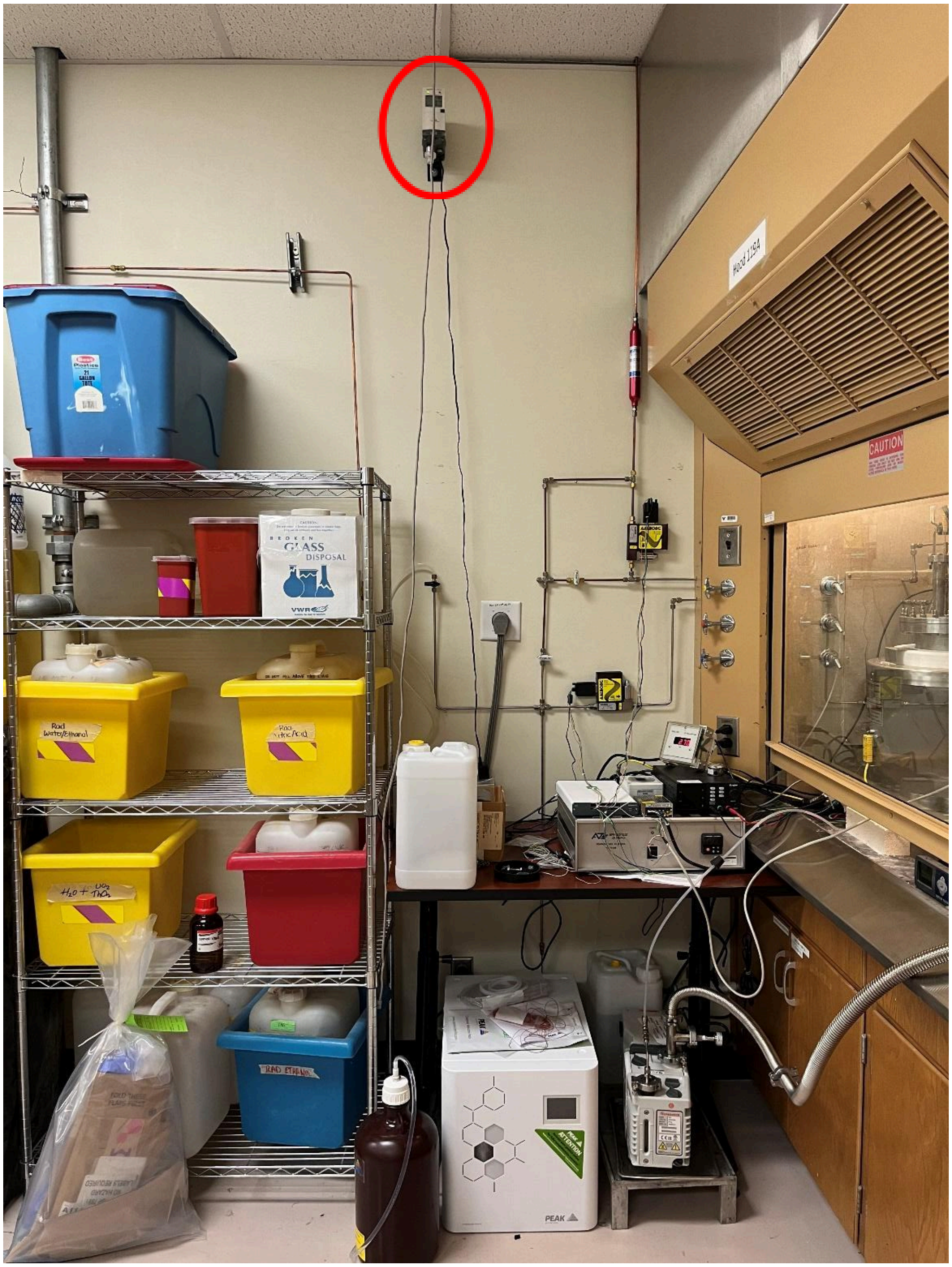
5.6.1. Close the Labview application.
5.6.2. Ensure that all RGA data has been saved to a separate file, then close the RGA application.
5.6.3. Close the Leyassist application.
5.6.4. Ensure that all valves on the system are closed.
5.6.5. Ensure that the high vacuum pressure gauge is turned off.
5.6.6. Ensure that both vacuum pumps are turned off.

5.7. Containment Loss

Although every precaution has been taken by the PI to minimize the chance of this happening, accidental loss of containment within the system, particularly during the Fluorination step, presents an acute and large health risk to the operator and others present in the lab. The fume hood provides an extra layer of containment for those at risk, but steps should be taken to minimize the health risk to operators and contamination of products.

5.7.1. The HF detector (shown below) is programmed to sound an alarm should the HF concentration in the fume hood exceed 3 ppm. This will also cause the program to automatically close the HF flow controller and purge the system with argon for several minutes and call the PI with an alarm message. The operator should evacuate all personnel from the room and wait until the detector reads below 3 ppm before shutting the regulator valve and cylinder valve on the HF cylinder. The PI should then be contacted to assess the leak location and fix it before restarting the system.

5.7.2. The H2 detector (shown below) is programmed to sound an alarm should the hydrogen concentration in the room exceed 4% by volume. The operator should press the pause button in the Labview program before stopping and shutting off the hydrogen generator. The PI should then be contacted to assess the leak location and fix it before restarting the system.



5.8. Cleaning of Reactor and Transfer Vessels

If the type of salt is changed or other changes are made to the process, the transfer vessel, reaction vessel liner, and the dip tubes will need to be cleaned to minimize contamination. All liquid waste produced by cleaning must be put into an approved bottle.

5.9.1. For cleaning the transfer vessel

a) Rinse the vessel and lid with ASTM Type I water.
b) Soak vessel and lid in 0.16 M (1 wt%) HNO_3 solution at 90°C ± 10°C for 1 hr.
c) Dispose of the HNO_3 solution into a proper waste container.
d) Rinse the vessel and lid again with ASTM Type I water.
e) Soak the vessel and lid in ASTM Type 1 water at 90°C ± 10°C for 1 hr.
f) Fill the vessel 90% full of ASTM Type 1 water and put on the lid.
g) Place the filled vessel into an oven at 90°C ± 10°C for 16 hr.
h) Remove the water from the vessel and dry in an oven at 90°C ± 10°C for 16 hr.
i) Connect argon line to the vessel inlet and open the cylinder slightly for continuous argon flow.
j) Place the vessel into a 110°C ± 10°C oven for 1 hr.
k) Allow the vessel to cool to room temperature.
l) Close the cylinder, then the transfer vessel feed valve.
m) Remove argon line. The vessel should now be clean and purged of all moisture and air.

5.9.2. For cleaning the reaction vessel liner and dip tubes

a) Rinse the liner and dip tubes with ASTM Type I water.
b) Soak the vessel and lid in ASTM Type 1 water at 90°C ± 10°C for 1 hr.
c) Fill the liner 90% full of ASTM Type 1 water and put on the lid.
d) Place the filled liner and the dip tubes into an oven at 90°C ± 10°C for 16 hr.
e) Remove the water from the liner and dry in an oven at 90°C ± 10°C for another 16 hr.
f) The liner and dip tubes should now be clean.

5.9. Replacing Scrubber Reagent

Both the wet and dry scrubber reagents need to be replaced either between purifications or when the material appears to be consumed. For the wet scrubber, this will be when the reagent turns pink in tint/color. The dry scrubber reagent needs to be replaced at the same time as the wet scrubber reagent. Ensure gloves, lab coats, and safety glasses are worn when performing this task.

5.9.1. For the wet scrubbers

a) Pull the dip tube out of the carboy and rinse with distilled water.
b) Seal the carboy using the included cap.
c) Secure a hazardous waste tag around the handle of the carboy.
d) Place the carboy on the hazardous waste shelf or cart.
e) Replace with a fresh carboy.
f) Fill the new carboy roughly 3/4 of the way with cool tap water.
g) Use a funnel to slowly add roughly 500 g of KOH to the carboy.
h) Insert dip tube into the carboy.

5.9.2. For the dry scrubber

a) Remove the cap from the bottle by screwing the bottle away from the cap.
b) Empty its contents into a proper disposal container.
c) Secure a hazardous waste tag around the container.
d) Place the container on the hazardous waste shelf or cart.
e) Refill the bottle 2/3 with dry CaO.
f) Put the cap back on the bottle by screwing the bottle into the cap.
g) Shake the bottle to ensure the dip tubes go beneath the surface of the CaO.

5.10. Fuel/Fission Product Addition

Currently, no depleted uranium or thorium fuel, or surrogate fission products are added to the purified salt to simulate reactor conditions. If this is requested, this SOP will be revised to include this process.